

IE212 – CÔNG NGHỆ DỮ LIỆU LỚN

LAB 1: HADOOP, HDFS & MAPREDUCE

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Lớp: IE212.R12.1

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B. Thực hành tại lớp

B.1 TransAnalysis1: tổng tiền theo loại hình thể thao

Ảnh chụp hadoop jar:

```
23521720_NguyenMinhTuan:~/Lab01$ hadoop jar /home/cloudera/Lab01/lab1.jar TransAnalysis1 /user/cloudera/lab1/trans /user/cloudera/lab1/out1
26/10/05 08:45:40 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/10/05 08:45:41 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
26/10/05 08:45:42 INFO input.FileInputFormat: Total input paths to process : 1
26/10/05 08:45:42 INFO mapreduce.JobSubmitter: number of splits:1
26/10/05 08:45:42 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1791210122571_0001
26/10/05 08:45:44 INFO impl.YarnClientImpl: Submitted application application_1791210122571_0001
26/10/05 08:45:44 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1791210122571_0001/
26/10/05 08:45:44 INFO mapreduce.Job: Running job: job_1791210122571_0001
26/10/05 08:45:56 INFO mapreduce.Job: Job job_1791210122571_0001 running in uber mode : false
26/10/05 08:45:56 INFO mapreduce.Job: map 0% reduce 0%
26/10/05 08:46:09 INFO mapreduce.Job: map 100% reduce 0%
26/10/05 08:46:16 INFO mapreduce.Job: map 100% reduce 100%
26/10/05 08:46:16 INFO mapreduce.Job: Job job_1791210122571_0001 completed successfully
26/10/05 08:46:16 INFO mapreduce.Job: Counters: 49
```

Ảnh chụp hdfs dfs -cat:

```
23521720_NguyenMinhTuan:~/Lab01$ hdfs dfs -cat /user/cloudera/lab1/out1/part-r-000000
Air Sports      45.790000915527344
Combat Sports  169.2199993133545
Exercise & Fitness  569.4400081634521
Games          256.6899962425232
Gymnastics     386.7900047302246
Indoor Games   213.82999801635742
Jumping        352.9800033569336
Outdoor Play Equipment  536.6800003051758
Outdoor Recreation  1373.9800033569336
Puzzles        27.889999389648438
Team Sports    694.8900051116943
Water Sports   891.0399971008301
Winter Sports  31.869999885559082
23521720_NguyenMinhTuan:~/Lab01$
```

B.2. TransAnalysis2: kèm danh sách người chơi

Ảnh chụp hadoop jar:

```
23521720_NguyenMinhTuan:~/Lab01$ hadoop jar /home/cloudera/Lab01/lab1.jar TransAnalysis2 /user/cloudera/lab1/trans /user/cloudera/lab1/out2
26/10/05 08:53:36 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/10/05 08:53:37 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
26/10/05 08:53:37 INFO input.FileInputFormat: Total input paths to process : 1
26/10/05 08:53:37 INFO mapreduce.JobSubmitter: number of splits:1
26/10/05 08:53:37 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1791210122571_0002
26/10/05 08:53:38 INFO impl.YarnClientImpl: Submitted application application_1791210122571_0002
26/10/05 08:53:38 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1791210122571_0002/
26/10/05 08:53:38 INFO mapreduce.Job: Running job: job_1791210122571_0002
26/10/05 08:53:45 INFO mapreduce.Job: Job job_1791210122571_0002 running in uber mode : false
26/10/05 08:53:45 INFO mapreduce.Job: map 0% reduce 0%
26/10/05 08:53:50 INFO mapreduce.Job: map 100% reduce 0%
26/10/05 08:53:56 INFO mapreduce.Job: map 100% reduce 100%
26/10/05 08:53:56 INFO mapreduce.Job: Job job_1791210122571_0002 completed successfully
26/10/05 08:53:56 INFO mapreduce.Job: Counters: 49
```

Ảnh chụp hdfs dfs -cat:

```
23521720_NguyenMinhTuan:~/Lab01$ hdfs dfs -cat /user/cloudera/lab1/out2/part-r-00000
Air Sports [4000005] 45.790000915527344
Combat Sports [4000001,4000009] 169.2199993133545
Exercise & Fitness [4000010,4000005,4000001,4000007,4000001,4000002,4000002] 569.4400081634521
Games [4000008,4000010,4000008] 256.6899962425232
Gymnastics [4000003,4000009,4000001,4000010] 386.7900047302246
Indoor Games [4000009,4000004,4000009] 213.82999801635742
Jumping [4000010,4000006,4000010] 352.9800033569336
Outdoor Play Equipment [4000005,4000009,4000008,4000006,4000006] 536.6800003051758
Outdoor Recreation [4000001,4000003,4000002,4000007,4000004,4000007,4000007,4000008,4000007,4000004] 1373.9800033569336
Puzzles [4000005] 27.889999389648438
Team Sports [4000005,4000008,4000008,4000008,4000002,4000008,4000007,4000010] 694.8900051116943
Water Sports [4000002,4000001,4000008,4000008,4000004,4000001,4000002,4000004,4000006,4000003,4000009] 891.0399971008301
Winter Sports [4000001,4000006] 31.869999885559082
23521720_NguyenMinhTuan:~/Lab01$ S
```

B.3. TransAnalysis3: bỏ các ID trùng

Ảnh chụp hadoop jar:

```
23521720_NguyenMinhTuan:~/Lab01$ hadoop jar /home/cloudera/Lab01/lab1.jar TransAnalysis3 /user/cloudera/lab1/trans /user/cloudera/lab1/out3
26/10/05 08:58:01 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/10/05 08:58:01 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
26/10/05 08:58:02 INFO input.FileInputFormat: Total input paths to process : 1
26/10/05 08:58:02 WARN hdfs.DFSClient: Caught exception
java.lang.InterruptedException
    at java.lang.Object.wait(Native Method)
    at java.lang.Thread.join(Thread.java:1281)
    at java.lang.Thread.join(Thread.java:1355)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.closeResponder(DFSOutputStream.java:967)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.endBlock(DFSOutputStream.java:705)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.run(DFSOutputStream.java:894)
26/10/05 08:58:02 INFO mapreduce.JobSubmitter: number of splits:1
26/10/05 08:58:02 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1791210122571_0003
26/10/05 08:58:02 INFO impl.YarnClientImpl: Submitted application application_1791210122571_0003
26/10/05 08:58:02 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1791210122571_0003/
26/10/05 08:58:02 INFO mapreduce.Job: Running job: job_1791210122571_0003
26/10/05 08:58:10 INFO mapreduce.Job: Job job_1791210122571_0003 running in uber mode : false
26/10/05 08:58:10 INFO mapreduce.Job: map 0% reduce 0%
26/10/05 08:58:17 INFO mapreduce.Job: map 100% reduce 0%
26/10/05 08:58:24 INFO mapreduce.Job: map 100% reduce 100%
26/10/05 08:58:24 INFO mapreduce.Job: Job job_1791210122571_0003 completed successfully
26/10/05 08:58:24 INFO mapreduce.Job: Counters: 49
```

Ảnh chụp hdfs dfs -cat:

```
23521720_NguyenMinhTuan:~/Lab01$ hdfs dfs -cat /user/cloudera/lab1/out3/part-r-00000
Air Sports [4000005] 45.790000915527344
Combat Sports [4000001,4000009] 169.2199993133545
Exercise & Fitness [4000010,4000005,4000001,4000007,4000002] 569.4400081634521
Games [4000008,4000010] 256.6899962425232
Gymnastics [4000003,4000009,4000001,4000010] 386.7900047302246
Indoor Games [4000009,4000004] 213.82999801635742
Jumping [4000010,4000006] 352.9800033569336
Outdoor Play Equipment [4000005,4000009,4000008,4000006] 536.6800003051758
Outdoor Recreation [4000001,4000003,4000002,4000007,4000004,4000008] 1373.9800033569336
Puzzles [4000005] 27.889999389648438
Team Sports [4000005,4000008,4000002,4000007,4000010] 694.8900051116943
Water Sports [4000002,4000001,4000008,4000004,4000006,4000003,4000009] 891.0399971008301
Winter Sports [4000001,4000006] 31.869999885559082
23521720_NguyenMinhTuan:~/Lab01$
```

B.4. TransAnalysis4: đếm thay vì liệt kê

Ảnh chụp hadoop jar:

```
23521720_NguyenMinhTuan:~/Lab01$ hadoop jar /home/cloudera/Lab01/lab1.jar TransAnalysis4 /user/cloudera/lab1/trans /user/cloudera/lab1/out4
26/10/05 09:02:37 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/10/05 09:02:37 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
26/10/05 09:02:37 WARN hdfs.DFSClient: Caught exception
java.lang.InterruptedException
    at java.lang.Object.wait(Native Method)
    at java.lang.Thread.join(Thread.java:1281)
    at java.lang.Thread.join(Thread.java:1355)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.closeResponder(DFSOutputStream.java:967)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.endBlock(DFSOutputStream.java:705)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.run(DFSOutputStream.java:894)
26/10/05 09:02:37 INFO input.FileInputFormat: Total input paths to process : 1
26/10/05 09:02:38 INFO mapreduce.JobSubmitter: number of splits:1
26/10/05 09:02:38 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1791210122571_0004
26/10/05 09:02:38 INFO impl.YarnClientImpl: Submitted application application_1791210122571_0004
26/10/05 09:02:38 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1791210122571_0004/
26/10/05 09:02:38 INFO mapreduce.Job: Running job: job_1791210122571_0004
26/10/05 09:02:45 INFO mapreduce.Job: Job job_1791210122571_0004 running in uber mode : false
26/10/05 09:02:45 INFO mapreduce.Job:  map 0% reduce 0%
26/10/05 09:02:50 INFO mapreduce.Job:  map 100% reduce 0%
26/10/05 09:02:57 INFO mapreduce.Job:  map 100% reduce 100%
26/10/05 09:02:58 INFO mapreduce.Job: Job job_1791210122571_0004 completed successfully
26/10/05 09:02:59 INFO mapreduce.Job: Counters: 49
```

Ảnh chụp hdfs dfs -cat:

```
23521720_NguyenMinhTuan:~/Lab01$ hdfs dfs -cat /user/cloudera/lab1/out4/part-r-00000
Air Sports      1  45.790000915527344
Combat Sports  2  169.2199993133545
Exercise & Fitness  5  569.4400081634521
Games          2  256.6899962425232
Gymnastics     4  386.7900047302246
Indoor Games   2  213.82999801635742
Jumping        2  352.9800033569336
Outdoor Play Equipment  4  536.6800003051758
Outdoor Recreation  6  1373.9800033569336
Puzzles        1  27.889999389648438
Team Sports     5  694.8900051116943
Water Sports    7  891.0399971008301
Winter Sports   2  31.869999885559082
23521720_NguyenMinhTuan:~/Lab01$
```

B.5. TransAnalysis5: nối dữ liệu ở bước Map

Ảnh chụp hadoop jar:

```

23521720_NguyenMinhTuan:~/Lab01$ hadoop jar /home/cloudera/Lab01/lab1.jar TransAnalysis5 /user/cloudera/lab1/trans /user/cloudera/lab1/out5
26/10/05 09:06:04 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/10/05 09:06:04 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
26/10/05 09:06:05 INFO input.FileInputFormat: Total input paths to process : 1
26/10/05 09:06:05 WARN hdfs.DFSClient: Caught exception
java.lang.InterruptedException
    at java.lang.Object.wait(Native Method)
    at java.lang.Thread.join(Thread.java:1281)
    at java.lang.Thread.join(Thread.java:1355)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.closeResponder(DFSOutputStream.java:967)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.endBlock(DFSOutputStream.java:705)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.run(DFSOutputStream.java:894)
26/10/05 09:06:05 INFO mapreduce.JobSubmitter: number of splits:1
26/10/05 09:06:06 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1791210122571_0005
26/10/05 09:06:06 INFO impl.YarnClientImpl: Submitted application application_1791210122571_0005
26/10/05 09:06:06 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1791210122571_0005/
26/10/05 09:06:06 INFO mapreduce.Job: Running job: job_1791210122571_0005
26/10/05 09:06:19 INFO mapreduce.Job: Job job_1791210122571_0005 running in uber mode : false
26/10/05 09:06:19 INFO mapreduce.Job:  map 0% reduce 0%
26/10/05 09:06:31 INFO mapreduce.Job:  map 100% reduce 0%
26/10/05 09:06:40 INFO mapreduce.Job:  map 100% reduce 100%
26/10/05 09:06:41 INFO mapreduce.Job: Job job_1791210122571_0005 completed successfully
26/10/05 09:06:42 INFO mapreduce.Job: Counters: 49

```

Ảnh chụp hdfs dfs -cat:

```

23521720_NguyenMinhTuan:~/Lab01$ hdfs dfs -cat /user/cloudera/lab1/out5/part-r-00000
Air Sports      [Karen]      45.790000915527344
Combat Sports  [Kristina,Malcolm]  169.2199993133545
Exercise & Fitness  [Dolores,Karen,Kristina,Elsie,Paige]  569.4400081634521
Games          [Hazel,Dolores]  256.6899962425232
Gymnastics     [Sherri,Malcolm,Kristina,Dolores]  386.7900047302246
Indoor Games   [Malcolm,Gretchen]  213.82999801635742
Jumping        [Dolores,Patrick]  352.9800033569336
Outdoor Play Equipment  [Karen,Malcolm,Hazel,Patrick]  536.6800003051758
Outdoor Recreation  [Kristina,Sherri,Paige,Elsie,Gretchen,Hazel]  1373.9800033569336
Puzzles        [Karen]      27.889999389648438
Team Sports    [Karen,Hazel,Paige,Elsie,Dolores]  694.8900051116943
Water Sports   [Paige,Kristina,Hazel,Gretchen,Patrick,Sherri,Malcolm]  891.0399971008301
Winter Sports  [Kristina,Patrick]  31.869999885559082
23521720_NguyenMinhTuan:~/Lab01$ █

```

B.6. TransAnalysis6: thêm tuổi vào danh sách

Ảnh chụp hadoop jar:

```

23521720_NguyenMinhTuan:~/Lab01$ hadoop jar /home/cloudera/Lab01/lab1.jar TransAnalysis6 /user/cloudera/lab1/trans /user/cloudera/lab1/out6
26/10/05 09:17:04 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/10/05 09:17:04 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
26/10/05 09:17:04 WARN hdfs.DFSClient: Caught exception
java.lang.InterruptedException
    at java.lang.Object.wait(Native Method)
    at java.lang.Thread.join(Thread.java:1281)
    at java.lang.Thread.join(Thread.java:1355)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.closeResponder(DFSOutputStream.java:967)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.endBlock(DFSOutputStream.java:705)
    at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.run(DFSOutputStream.java:894)
26/10/05 09:17:04 INFO input.FileInputFormat: Total input paths to process : 1
26/10/05 09:17:05 INFO mapreduce.JobSubmitter: number of splits:1
26/10/05 09:17:05 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1791210122571_0006
26/10/05 09:17:05 INFO impl.YarnClientImpl: Submitted application application_1791210122571_0006
26/10/05 09:17:05 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1791210122571_0006/
26/10/05 09:17:05 INFO mapreduce.Job: Running job: job_1791210122571_0006
26/10/05 09:17:12 INFO mapreduce.Job: Job job_1791210122571_0006 running in uber mode : false
26/10/05 09:17:12 INFO mapreduce.Job:  map 0% reduce 0%
26/10/05 09:17:18 INFO mapreduce.Job:  map 100% reduce 0%
26/10/05 09:17:25 INFO mapreduce.Job:  map 100% reduce 100%
26/10/05 09:17:25 INFO mapreduce.Job: Job job_1791210122571_0006 completed successfully
26/10/05 09:17:25 INFO mapreduce.Job: Counters: 49

```

Ảnh chụp hdfs dfs -cat:

```
23521720_NguyenMinhTuan:~/Lab01$ hdfs dfs -cat /user/cloudera/lab1/out6/part-r-00000
Air Sports      [Karen,74] 45.790000915527344
Combat Sports   [Kristina,55,Malcolm,39] 169.2199993133545
Exercise & Fitness [Dolores,60,Karen,74,Kristina,55,Elsie,43,Paige,74] 569.4400081634521
Games           [Hazel,63,Dolores,60] 256.6899962425232
Gymnastics      [Sherri,34,Malcolm,39,Kristina,55,Dolores,60] 386.7900047302246
Indoor Games    [Malcolm,39,Gretchen,66] 213.82999801635742
Jumping         [Dolores,60,Patrick,42] 352.9800033569336
Outdoor Play Equipment [Karen,74,Malcolm,39,Hazel,63,Patrick,42] 536.6800003051758
Outdoor Recreation [Kristina,55,Sherri,34,Paige,74,Elsie,43,Gretchen,66,Hazel,63] 1373.9800033569336
Puzzles         [Karen,74] 27.889999389648438
Team Sports     [Karen,74,Hazel,63,Paige,74,Elsie,43,Dolores,60] 694.8900051116943
Water Sports    [Paige,74,Kristina,55,Hazel,63,Gretchen,66,Patrick,42,Sherri,34,Malcolm,39] 891.0399971008301
Winter Sports   [Kristina,55,Patrick,42] 31.869999885559082
23521720_NguyenMinhTuan:~/Lab01$
```

C. Bài tập về nhà

C.1. Bài tập bắt buộc

C.1.1. Bài 1. Điểm trung bình và số lượt đánh giá của từng phim

Ảnh chụp hadoop jar:

```
23521720_NguyenMinhTuan:~/Lab01$ hadoop jar /home/cloudera/Lab01/lab1.jar MovieRatingAverage /user/cloudera/lab1/ml/ratings_1.txt /user/cloudera/lab1/ml/ratings_2.txt /user/cloudera/lab1/cl_bail
26/10/05 09:23:11 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/10/05 09:23:11 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
26/10/05 09:23:12 INFO input.FileInputFormat: Total input paths to process : 2
26/10/05 09:23:12 INFO mapreduce.JobSubmitter: number of splits:2
26/10/05 09:23:12 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1791210122571_0008
26/10/05 09:23:12 INFO impl.YarnClientImpl: Submitted application application_1791210122571_0008
26/10/05 09:23:12 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1791210122571_0008/
26/10/05 09:23:12 INFO mapreduce.Job: Running job: job_1791210122571_0008
26/10/05 09:23:18 INFO mapreduce.Job: Job job_1791210122571_0008 running in uber mode : false
26/10/05 09:23:18 INFO mapreduce.Job: map 0% reduce 0%
26/10/05 09:23:24 INFO mapreduce.Job: map 50% reduce 0%
26/10/05 09:23:25 INFO mapreduce.Job: map 100% reduce 0%
26/10/05 09:23:30 INFO mapreduce.Job: map 100% reduce 100%
26/10/05 09:23:30 INFO mapreduce.Job: Job job_1791210122571_0008 completed successfully
26/10/05 09:23:30 INFO mapreduce.Job: Counters: 49
```

Ảnh chụp hdfs dfs -cat:

```

23521720_NguyenMinhTuan:~/Lab01$ hdfs dfs -cat /user/cloudera/lab1/c1_bail/part-r-00000
Toy Story (1995) 4.25 (2)
The Shawshank Redemption (1994) 4.00 (2)
Back to the Future (1985) 4.00 (2)
Gladiator (2000) 3.25 (2)
Braveheart (1995) 4.25 (2)
Jurassic Park (1993) 4.00 (2)
The Departed (2006) 3.25 (2)
The Silence of the Lambs (1991) 3.50 (2)
Saving Private Ryan (1998) 4.00 (2)
Avatar (2009) 4.75 (2)
Interstellar (2014) 3.50 (2)
The Prestige (2006) 4.00 (2)
The Avengers (2012) 3.75 (2)
Memento (2000) 4.00 (2)
The Green Mile (1999) 3.50 (2)
The Matrix (1999) 3.75 (2)
The Lord of the Rings: The Fellowship of the Ring (2001) 4.25 (2)
The Lord of the Rings: The Two Towers (2002) 3.75 (2)
The Lord of the Rings: The Return of the King (2003) 3.50 (2)
Fight Club (1999) 4.25 (2)
Se7en (1995) 3.75 (2)
The Usual Suspects (1995) 4.25 (2)
Schindler's List (1993) 4.25 (2)
American Beauty (1999) 3.75 (2)
Good Will Hunting (1997) 3.50 (2)
The Social Network (2010) 3.75 (2)
Slumdog Millionaire (2008) 4.25 (2)
Whiplash (2014) 3.75 (2)
La La Land (2016) 4.00 (2)
Inception (2010) 3.75 (2)
Mad Max: Fury Road (2015) 4.00 (2)
The Revenant (2015) 3.50 (2)
The Dark Knight (2008) 3.50 (2)
Gravity (2013) 4.25 (2)
The Wolf of Wall Street (2013) 4.25 (2)
Birdman (2014) 3.00 (2)
Pulp Fiction (1994) 4.00 (2)
The Big Short (2015) 4.50 (2)
Manchester by the Sea (2016) 4.00 (2)
Moonlight (2016) 3.50 (2)
Spotlight (2015) 4.25 (2)
The Imitation Game (2014) 3.75 (2)
The Assassination of Jesse James (2017) 3.75 (2)
Forrest Gump (1994) 3.75 (2)
Dunkirk (2017) 3.50 (2)
Coco (2017) 4.25 (2)
Parasite (2019) 4.00 (2)
The Godfather (1972) 3.50 (2)
Titanic (1997) 3.50 (2)
Star Wars: A New Hope (1977) 3.50 (2)
The Lion King (1994) 4.00 (2)
Highest Rated Movie Avatar (2009) is the highest rated movie with an average rating of 4.75 among movies with at least 2 ratings.
23521720_NguyenMinhTuan:~/Lab01$ █

```

C.1.2. Bài 2. Điểm trung bình theo thể loại phim

Ảnh chụp hadoop jar:

```

23521720 NguyenMinhTuan:~/Lab01$ hadoop jar /home/cloudera/Lab01/lab1.jar GenreRatingAverage /user/cloudera/lab1/ml/ratings_1.txt /user/cloudera/lab1/ml/ratings_2.txt /user/cloudera/lab1/c1_bai2
26/10/05 09:25:44 INFO client.RMPProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/10/05 09:25:44 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
26/10/05 09:25:44 INFO input.FileInputFormat: Total input paths to process : 2
26/10/05 09:25:44 INFO mapreduce.JobSubmitter: number of splits:2
26/10/05 09:25:44 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1791210122571_0009
26/10/05 09:25:45 INFO impl.YarnClientImpl: Submitted application application_1791210122571_0009
26/10/05 09:25:45 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1791210122571_0009/
26/10/05 09:25:45 INFO mapreduce.Job: Running job: job_1791210122571_0009
26/10/05 09:25:51 INFO mapreduce.Job: Job job_1791210122571_0009 running in uber mode : false
26/10/05 09:25:51 INFO mapreduce.Job: map 0% reduce 0%
26/10/05 09:26:00 INFO mapreduce.Job: map 50% reduce 0%
26/10/05 09:26:01 INFO mapreduce.Job: map 100% reduce 0%
26/10/05 09:26:05 INFO mapreduce.Job: map 100% reduce 100%
26/10/05 09:26:05 INFO mapreduce.Job: Job job_1791210122571_0009 completed successfully
26/10/05 09:26:05 INFO mapreduce.Job: Counters: 49

```

Ảnh chụp hdfs dfs -cat:

```

23521720 NguyenMinhTuan:~/Lab01$ hdfs dfs -cat /user/cloudera/lab1/c1_bai2/part-r-000000
Action 3.73 (20)
Adventure 3.85 (30)
Animation 4.17 (6)
Biography 4.14 (14)
Children 4.25 (2)
Comedy 3.96 (12)
Crime 3.78 (20)
Drama 3.81 (76)
Family 4.25 (2)
Fantasy 3.95 (10)
History 4.00 (6)
Music 3.88 (4)
Mystery 4.00 (8)
Romance 3.75 (8)
Sci-Fi 3.85 (20)
Thriller 3.88 (16)
War 4.00 (2)
23521720_NguyenMinhTuan:~/Lab01$

```

C.1.3. Bài 3. Điểm trung bình theo giới tính

Ảnh chụp hadoop jar:

```

23521720 NguyenMinhTuan:~/Lab01$ hadoop jar /home/cloudera/Lab01/lab1.jar GenderRatingAverage /user/cloudera/lab1/ml/movies.txt /user/cloudera/lab1/ml/ratings_1.txt /user/cloudera/lab1/ml/ratings_2.txt /user/cloudera/lab1/c1_bai3
26/10/05 09:27:08 INFO client.RMPProxy: Connecting to ResourceManager at /0.0.0.0:8032
26/10/05 09:27:08 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
26/10/05 09:27:09 INFO input.FileInputFormat: Total input paths to process : 1
26/10/05 09:27:09 INFO input.FileInputFormat: Total input paths to process : 2
26/10/05 09:27:09 INFO mapreduce.JobSubmitter: number of splits:3
26/10/05 09:27:09 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1791210122571_0010
26/10/05 09:27:09 INFO impl.YarnClientImpl: Submitted application application_1791210122571_0010
26/10/05 09:27:09 INFO mapreduce.Job: The url to track the job: http://quickstart.cloudera:8088/proxy/application_1791210122571_0010/
26/10/05 09:27:09 INFO mapreduce.Job: Running job: job_1791210122571_0010
26/10/05 09:27:15 INFO mapreduce.Job: Job job_1791210122571_0010 running in uber mode : false
26/10/05 09:27:15 INFO mapreduce.Job: map 0% reduce 0%
26/10/05 09:27:25 INFO mapreduce.Job: map 33% reduce 0%
26/10/05 09:27:27 INFO mapreduce.Job: map 100% reduce 0%
26/10/05 09:27:31 INFO mapreduce.Job: map 100% reduce 100%
26/10/05 09:27:31 INFO mapreduce.Job: Job job_1791210122571_0010 completed successfully
26/10/05 09:27:31 INFO mapreduce.Job: Counters: 49

```

Ảnh chụp hdfs dfs -cat:

```
23521720_NguyenMinhTuan:~/Lab01$ hdfs dfs -cat /user/cloudera/lab1/c1_bai3/part-r-00000
Toy Story (1995)      4.50, 4.00
The Shawshank Redemption (1994) 3.00, 5.00
Back to the Future (1985) 4.00, 4.00
Gladiator (2000)     3.50, 3.00
Braveheart (1995)    4.50, 4.00
Jurassic Park (1993) 3.50, 4.50
The Departed (2006)  4.50, 2.00
The Silence of the Lambs (1991) 3.50, 3.50
Saving Private Ryan (1998) 4.00, 4.00
Avatar (2009)        5.00, 4.50
Interstellar (2014)  3.50, 3.50
The Prestige (2006)  4.00, 4.00
The Avengers (2012)  3.50, 4.00
Memento (2000)       3.50, 4.50
The Green Mile (1999) 3.00, 4.00
The Matrix (1999)     3.00, 4.50
The Lord of the Rings: The Fellowship of the Ring (2001) 4.50, 4.00
The Lord of the Rings: The Two Towers (2002) 4.50, 3.00
The Lord of the Rings: The Return of the King (2003) 4.00, 3.00
Fight Club (1999)     5.00, 3.50
Se7en (1995)          4.00, 3.50
The Usual Suspects (1995) 4.00, 4.50
Schindler's List (1993) 4.00, 4.50
American Beauty (1999) 3.00, 4.50
Good Will Hunting (1997) 3.00, 4.00
The Social Network (2010) 3.50, 4.00
Slumdog Millionaire (2008) 4.50, 4.00
Whiplash (2014)       4.50, 3.00
La La Land (2016)     5.00, 3.00
Inception (2010)      3.50, 4.00
Mad Max: Fury Road (2015) 4.00, 4.00
The Revenant (2015)   3.50, 3.50
The Dark Knight (2008) 4.00, 3.00
Gravity (2013)        4.00, 4.50
The Wolf of Wall Street (2013) 4.00, 4.50
Birdman (2014)        3.00, 3.00
Pulp Fiction (1994)   4.50, 3.50
The Big Short (2015)  5.00, 4.00
Manchester by the Sea (2016) 3.50, 4.50
Moonlight (2016)      3.00, 4.00
Spotlight (2015)      4.50, 4.00

The Imitation Game (2014) 4.50, 3.00
Forrest Gump (1994)      2.50, 5.00
Dunkirk (2017)          4.00, 3.00
Coco (2017)            5.00, 3.50
Parasite (2019)        4.00, 4.00
The Godfather (1972)    4.50, 2.50
Titanic (1997)          4.00, 3.00
Star Wars: A New Hope (1977) 4.00, 3.00
The Lion King (1994)    4.50, 3.50
23521720_NguyenMinhTuan:~/Lab01$
```